

AMENDMENT NO. 5 JUNE 2026
TO
IS 3196 (PART 1) : 2013 WELDED LOW CARBON STEEL CYLINDERS
EXCEEDING 5 LITRES WATER CAPACITY FOR LOW PRESSURE
LIQUEFIABLE GASES
PART 1 CYLINDERS FOR LIQUEFIED PETROLEUM GASES (LPG) —
SPECIFICATION
(Sixth Revision)

(Page 6, clause 10.2) — Substitute the following for the existing:

‘10.2 Foot Ring

The foot ring, where fitted as a separate fixture to the bottom end of the cylinder, shall be at least 20 mm away from the circumferential weld. The thickness of the sheet from which the foot ring is made shall not be less than the calculated wall thickness of the cylinder body. The foot ring may be intermittently welded with body of the cylinder. In case, the bottom edge is curled, the curling shall be inwards to facilitate safe handling. It shall be provided with holes for ventilation and if curled, drainage holes shall be provided to avoid corrosion. However, drainage holes may not be provided in case the curl is sealed by welding. The maximum permissible deviation from the vertical shall not exceed 1°. Foot rings shall be sufficiently strong and made of steel (*see 4.3*) or any other material as approved by statutory authority. The bottom of the foot ring shall not be less than 8 mm below the outside bottom of the cylinder shell for cylinders up to 34 litre nominal water capacity. For cylinders of more than 34 litre nominal water capacity and up to 50 litre nominal water capacity, this value shall be minimum 15 mm and for cylinders exceeding 50 litre nominal water capacity, this value shall be minimum 25 mm.

NOTE — Attachment such as VP (valve protection) shroud or foot ring, when directly welded to pressure retaining parts shall be designed to prevent crevice corrosion. They may have intermittent contact, such as scalloping or attachment by ears. However, they ought to be continuously welded at all points of contact with the pressure retaining body of the cylinder.’

[Page 8, clause 17.2.3(e)] — Substitute the following for the existing:

‘e) Volumetric expansion:

The ratio of volumetric expansion of the cylinder at burst to its initial volume shall be greater than or equal to the following values:

- 1) Where the minimum value of tensile strength is lower than 410 N/mm²:
 - i) 20 percent if the length of the cylinder is greater than its diameter; and
 - ii) 14 percent if the length of the cylinder is equal to or less than its diameter.
- 2) Where the minimum value of tensile strength is equal to or greater than 410 N/mm²:
 - i) 15 percent if the length of the cylinder is greater than its diameter; and
 - ii) 10 percent if the length of the cylinder is equal to or less than its diameter.’

(Page 8, clause 20.2) — Insert the following at the end:

‘k) Permanent and tamper proof marking in form of bar code (as approved by statutory authority).

NOTE — ‘Bar code’ means a machine-readable graphical picture having alphabet, character, number embedded in it and includes radio frequency identification or quick response code.’